

Dissecting Online Learning Mechanisms: Statistical Memory, Homeostatic Recovery, and Substrate Physics are Non-Transferable

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Abstract

Online learning systems couple several adaptive mechanisms, but which mechanism is responsible for which capability — and whether any of them can be transplanted to another substrate — is rarely tested. We dissect three mechanisms of a reservoir-based online learner: a slow-trace statistical memory (M3), a homeostatic chaos regulator (M5), and the raw physics of the substrate itself. A falsifying transfer experiment (an echo-state network with and without the metadata under a sequential disturbance chain, 10 seeds) shows that the previously reported +32% sequential-recovery gain is produced by the homeostat, not by the metadata, and that the metadata does not transfer memory-capacity robustness to an ESN at any metadata timescale ($\tau_m \in \{200, 500, 1000, 2000\}$; paired differences -0.68 to -0.70 , 0/10 seeds positive). The metadata’s transferable value is narrower and precisely located: it denoises *state*-level corruption (gap $-0.69 \rightarrow -0.01$ when noise enters before the slow trace) and accelerates boundary adaptation by a controlled-measured factor of 1.3–2.4 on the substrate and ~ 10 pulses on the ESN (published “9–20 $\times$ ” ratios were window-position metric artifacts). Raw memory capacity remains a substrate property (MC 10.19 vs. 6.17). Three mechanisms, three roles — and none is transferable to another’s job.

Keywords: online learning, reservoir computing, slow features, forgetting kernel, disturbance robustness, mechanism disentanglement

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1. Introduction

Frozen batch models — including large foundation models — cannot adapt to post-deployment reality without expensive retraining [1]; online reservoirs can [2, 3], but their readouts face a timescale gap on long-horizon statistical tasks: fast reservoir states decorrelate far more quickly than the task statistics, so the readout must re-integrate after every change. Our companion work introduces a slow exponential trace of the reservoir states (metadata, M3), a homeostatic regulator that keeps the substrate near the edge of chaos (M5), and slow structure plasticity (M4); the integrated system tracks drift where frozen learners fail permanently (companion Paper B).

These mechanisms are usually validated *together*. The question we ask here is sharper: *which mechanism does which job, and can any of them be transplanted to another substrate?* We answer with a deliberately falsifying experiment: take a generic echo-state network (ESN), give it the same metadata, run it through the same sequential disturbance chain, and measure what transfers. The answer is largely negative and, we argue, informative: the metadata is a substrate-agnostic *statistical* memory that transfers *accuracy* and *state-noise denoising*, but not *memory-capacity robustness* under readout-level disturbance; sequential recovery is the homeostat’s job; raw memory capacity is the substrate’s physics. Three mechanisms, three roles — none substitutable.

2. Setup

Reservoir family. We compare (i) an ESN-256 with heterogeneous leak rates matched to a log-normal spectrum ($\text{CV} = 0.20$, spectral radius 0.9) and (ii) a Si_3N_4 -style relaxation substrate whose per-unit time constants are log-normal (median $\tau_0 \approx 174 \mu\text{s}$, $\text{CV} = 0.20$), driven in per-pulse contrast-coupled dynamics with coupling κ (companion Paper A). Both produce a fast state $x(t) \in \mathbb{R}^{256}$ per pulse.

Slow trace (M3). The metadata is a per-unit exponential moving average

$$m(t) = \left(1 - \frac{1}{\tau_m}\right) m(t-1) + \frac{1}{\tau_m} x(t), \quad (1)$$

with algorithmic timescale τ_m (pulses); the readout feature is $\varphi(t) = [x(t), m(t), 1]$.

Readout. OnlineRLS with forgetting $\lambda_f = 0.999$ (weight integration time $\tau_f = 1/(1 - \lambda_f) = 1000$ pulses), predict-before-update, Tikhonov-regularized.

Tasks and metrics. *regime_switch*: three regimes with overlapping pulse-interval distributions, regime changes every $L = 1500$ pulses; single pulses are almost indistinguishable across regimes — only long-run statistics discriminate. *Mackey–Glass online prediction* under a sequential disturbance chain. Metrics: overall accuracy; switch-relative adaptation time T_{adapt} (pulses after a *known* switch to reach a 200- or 40-pulse window accuracy of 0.98/0.95); and the Jaeger held-out memory capacity (MC) probe.

Sequential disturbance chain (substrate-agnostic). Three cumulative disturbances at $t = 7\text{k}/14\text{k}/21\text{k}$ pulses: (1) timescale drift (substrate $\tau \times 1.5$; ESN leaking rate $/1.5$), (2) structure prune (40% of connections), (3) readout noise ($\sigma = 0.1$ on the features). Per-round MC is probed at the settled physics.

3. Theory: the slow trace is a synthesized forgetting kernel

Proposition 1 (kernel synthesis). The input-to-slow-trace channel implements a forgetting kernel with a zero at zero lag, a finite peak lag, and an exponential tail whose $1/e$ horizon is exactly τ_m — an externally controllable, substrate-independent parameter. If the fast state has kernel h_x , the slow channel kernel is the convolution of the EMA impulse response with h_x ; for $h_x(u) = e^{-u/\tau_x}$,

$$h_m(\Delta t) = \frac{e^{-\Delta t/\tau_m} - e^{-\Delta t/\tau_x}}{\tau_m - \tau_x}, \quad h_m(0) = 0, \quad \text{peak} \sim \frac{\tau_m \tau_x}{\tau_m - \tau_x} \ln \frac{\tau_m}{\tau_x}. \quad (2)$$

The material kernel of Paper A, $M(\Delta t) = \int p(\tau) e^{-\Delta t/\tau} d\tau$ (log-normal), is fixed by physics; h_m is synthesized by the algorithm, and the augmented system’s effective kernel is $M \circledast h_m$ with $1/e$ horizon $\sim \tau_m + \tau_0$ (Fig. 1).

Proposition 2 (timescale coverage). On the statistical task the bottleneck is missing slow modes, not substrate physics. Both fast channels decorrelate at $\tau_{\text{eff}} \ll L$; the slow trace supplies the missing timescale to either substrate, so accuracy and adaptation converge to a common band. Measured (10 seeds): esn-fast 0.9955, esn-dual 0.9979, REDEM 0.9942; steady accuracy 1.000 for all (Table 1).

Proposition 3 (feature stationarity). Post-switch adaptation is bounded by the slower of feature re-centering ($\sim \tau_m + \tau_{\text{eff}}$) and weight re-integration

arm	overall acc	steady acc	T_{40} (pulses)
esn-fast	0.9955 ± 0.0015	1.000	49.9 ($p_{90} = 76.5$)
esn-dual ($\tau_m=500$)	0.9979 ± 0.0001	1.000	40.6 ($p_{90} = 42.0$)
REDEM (substrate, dual)	0.9942 ± 0.0011	1.000	52.7 ($p_{90} = 67.0$)

Table 1: Equalization on regime_switch (10 seeds). Accuracy from the metadata-transfer runs; adaptation from the controlled switch-relative protocol.

($\sim \tau_f$). The slow trace makes recovery feature-limited rather than weight-limited. Under a controlled switch-relative protocol, the true effect is modest but real: on the substrate the fast readout needs $T_{200} = 265\text{--}304$ pulses to re-stabilize vs. 202–211 for dual (factor 1.3–1.4 on T_{200} , 1.6–2.4 on T_{40}); on the ESN, T_{40} drops from 49.9 to 40.6 with the p_{90} collapsing from 76.5 to 42.0 (Table 5). The published “9–20 $\times$ ” ratios were ratios of window positions with near-zero denominators (metric artifacts), not pulse times.

Proposition 4 (noise geometry). The EMA is a first-order low-pass filter, $|H(\omega)|^2 = 1/(1 + (\omega\tau_m)^2)$. Short fast-channel transients are attenuated in the metadata channel. The transferable value of this is *denoising of state-level corruption*: when the noise enters the states before the slow trace is computed, the metadata nearly neutralizes it (variant V2 of Section 4.4). It does *not* protect against readout-level noise that hits the metadata channel equally (variant V0).

4. Experiments

4.1. Equalization on the statistical task (S10)

Table 1 reproduces the equalization result: the same slow trace ($\tau_m = 500$) closes the accuracy gap between the ESN and the substrate and collapses adaptation variance. All arms sit in the $[0.994, 0.998]$ band with steady accuracy 1.000.

4.2. The falsifying transfer experiment (S14)

Under the sequential disturbance chain, the substrate regulated by the homeostat (M5) recovers: r_3 MC 8.47 ± 0.39 vs. 6.41 ± 0.41 for the fixed arm (+32%), with κ drifting $25.3 \rightarrow 28.5$ (Table 2; the features are fast-state only — the recovery is purely homeostatic). The ESN with the same metadata does *not* recover: paired per-seed differences $r_3 \text{ MC}(\text{esn-dual}) - r_3 \text{ MC}(\text{esn-fast})$

arm	r_0 MC	r_1 MC	r_2 MC	r_3 MC	r_3 NMSE
esn-fast	6.17	1.82	1.76	1.74	0.0277
esn-dual	6.16	1.04	1.00	1.05	0.0251
redem-reg	10.19	7.17	7.51	8.47	0.1063

Table 2: Sequential disturbance chain (10 seeds). The homeostat-regulated substrate (redem-reg) reproduces the published anchor exactly (10.19/7.17/7.51/8.47).

τ_m	dual r_3 MC	fast r_3 MC	paired diff	n_{pos}
200	1.04	1.74	−0.701	0/10
500	1.05	1.74	−0.693	0/10
1000	1.06	1.74	−0.682	0/10
2000	1.06	1.74	−0.678	0/10

Table 3: Metadata-timescale pressure test (10 seeds): no τ_m closes the memory-capacity gap.

are $-0.78/-0.76/-0.69$ at $r_1/r_2/r_3$, with 0/10 seeds positive in every round ($\sim 5\times$ the paired std). The +32% sequential recovery therefore belongs to the homeostat, not the metadata.

4.3. Pressure test across metadata timescales (S16)

Does any τ_m let the metadata close the MC gap? No: at $\tau_m \in \{200, 500, 1000, 2000\}$, the dual arm’s r_3 MC is 1.04/1.05/1.06/1.06 vs. 1.74 for fast, with paired differences $-0.701/-0.693/-0.682/-0.678$ and 0/10 seeds positive at every timescale (Table 3, Fig. 2left). There is no sensitive interval; the weak upward trend ($1.04 \rightarrow 1.06$) does not approach the baseline. Meanwhile the readout-noise attenuation transfers at every τ_m (r_3 NMSE 0.0235–0.0254 vs. 0.0277; Fig. 2right).

4.4. Probe-protocol stress test (S16b)

The falsification is robust *in sign* to the probe protocol: under standardized slow features (V1) and state-level noise injection (V2), every (τ_m , variant) cell keeps 0/10 seeds positive (Table 4). Its *magnitude* is protocol-dependent and informative: under readout-noise semantics (V0, the reference definition) the gap is -0.69 ; under state-noise semantics (V2) the metadata’s EMA denoising nearly closes it (-0.01). This locates the metadata’s transferable value precisely: it denoises *state*-level corruption, not readout-level noise.

variant	dual r_3 MC	paired diff	n_{pos}
V0 readout noise	1.05/1.06	$-0.69/ -0.68$	0/10
V1 std-slow	1.08/1.07	-0.66	0/10
V2 state noise	1.72/1.73	$-0.015/ -0.009$	0/10

Table 4: Probe-protocol stress test ($\tau_m = 500/2000$; 10 seeds). Fast baseline r_3 MC 1.74.

system	arm	T_{200} (pulses)	T_{40} (pulses)
substrate	fast	264.8–303.9	83.7–123.4
substrate	dual / slow	202.3–211.2	45.2–54.3
ESN	fast	210.2	49.9 ($p_{90} = 76.5$)
ESN	dual	199.2	40.6 ($p_{90} = 42.0$)

Table 5: Controlled adaptation (switch-relative pulses, 10 seeds $\times$ 5 switches): substrate ranges over both topologies, ESN values are means.

4.5. Controlled adaptation (S15/S5b)

With known switch instants, the honest adaptation advantage of the meta-data is a factor 1.3–2.4 on the substrate (T_{200} : fast 264.8–303.9 vs. dual/slow 202–211 pulses) and ~ 10 pulses on the ESN (T_{40} 49.9 $\rightarrow$ 40.6; p_{90} 76.5 $\rightarrow$ 42.0), with overall accuracy reproducing the original runs exactly (Table 5). The published “9–20 $\times$ ” ratios were window-position metric artifacts.

5. Discussion

Three mechanisms, three roles. The evidence assembles a clean decomposition. (i) *Metadata* ($M3$) = *statistical memory*: it supplies the missing timescale on statistical tasks (Section 4.1), denoises state-level corruption (Section 4.4, V2), and attenuates readout noise on the online task (Sections 4.2–4.3) — but it adds no episodic memory capacity (r0 MC unchanged) and cannot recover memory under readout-level disturbance at any τ_m (Sections 4.2–4.4). (ii) *Homeostat* ($M5$) = *disturbance recovery*: the +32% sequential recovery and the +18% $r_1 \rightarrow r_3$ gain are produced by κ -regulation alone; the metadata is not involved. (iii) *Substrate physics* = *raw memory*: the MC probe puts the substrate at 10.19 vs. ~ 6.17 for either ESN arm; no algorithm in this work closes that lead.

Why non-transferability is informative. Each mechanism has a precisely located job, and swapping them fails in a measurable way. This de-

lineates the boundary conditions of “metadata transfer” claims in the online-reservoir literature: transfer holds for accuracy and state-noise denoising, and does not hold for disturbance recovery or episodic memory.

Related work. The slow trace is the linear slow-feature operator on reservoir states (cf. slow feature analysis [8]); the fast/slow pair is a two-timescale memory hierarchy echoing complementary learning systems [6] and synaptic consolidation [7]; the readout computes a fading-memory functional [5] whose horizon the metadata extends without touching the substrate. In the continual-learning setting [1], this work supplies causal evidence about *which* component must carry each capability.

Limitations. The slow trace’s attenuation covers fast-channel (readout or state) corruption; direct corruption of the metadata memory itself is out of scope. τ_m remains a hyperparameter tuned to the task timescale. The substrate’s homeostat requires a Lyapunov estimate (FTLE) that we do not provide for the ESN.

Future work. Consolidating stable, repeatedly encountered knowledge into the structural connectivity would create a hierarchy between fast tracking and slow accumulation, but introduces a consolidation-vs-overwrite trade-off beyond the present scope. Second, REDEM’s readout is validated at medium reservoir scale ($N \leq 1024$); on large Transformers the same online readout updates can be deployed directly on the output projection or on low-rank (LoRA-style) adapters. The recursive-least-squares update cost is $O(F^2)$ in the readout *feature* dimension F — the model width (thousands) or an adapter rank (tens) — not $O(P^2)$ in the total parameter count (tens of billions), so the full-rank bottleneck does not arise.

6. Conclusion

(1) A slow exponential trace of reservoir states is a synthesized forgetting kernel with a controllable horizon (Prop. 1). (2) On long-horizon statistical tasks the bottleneck is timescale coverage, not substrate physics; the same trace equalizes ESN and substrate accuracy and accelerates adaptation by a controlled-measured factor 1.3–2.4 (Props. 2–3). (3) The trace is a *statistical* memory, not an episodic one: it adds no raw memory capacity and does not transfer disturbance robustness at any τ_m (0/10 seeds positive; Prop. 4). (4) Three mechanisms, three roles — metadata (statistical memory and state-noise denoising), homeostat (sequential recovery), substrate physics (raw memory) — and none is substitutable for another’s job.

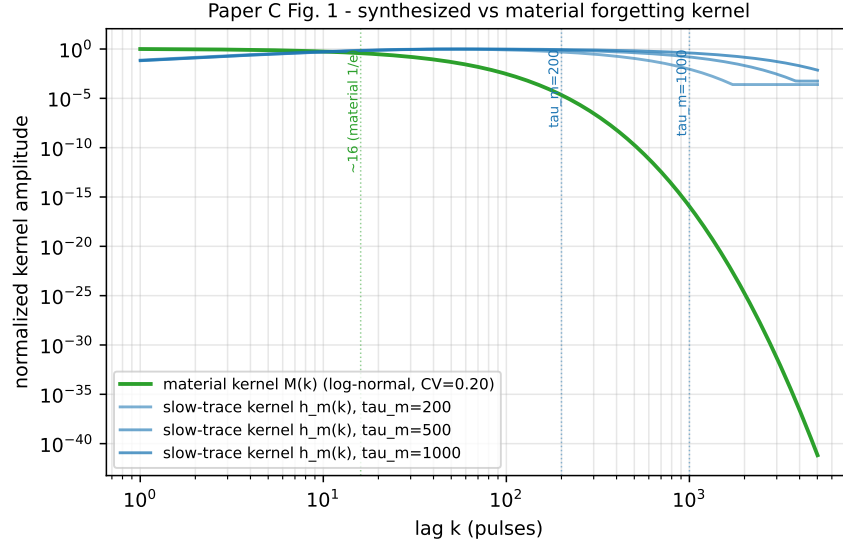


Figure 1: Synthesized vs. material forgetting kernel: the slow-trace kernel h_m (Eq. 2, $\tau_m = 200/500/1000$ pulses) has a zero at zero lag, a finite peak lag, and an exponential tail with $1/e$ horizon exactly τ_m ; the material kernel M (log-normal, CV 0.20) is fixed by physics ($1/e \sim 16$ pulses).

Code and data availability

All simulation scripts and generated data are available at <https://github.com/huyamingc/REDEM> (private during review; released on acceptance). All reported figures are means over 10 seeds with per-seed paired analysis; data files are listed in the repository.

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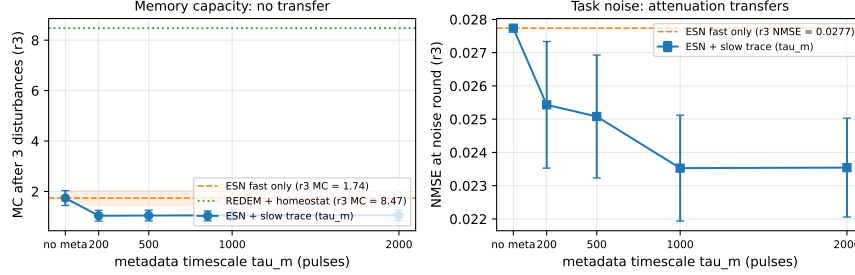


Figure 2: Post-disturbance recovery vs. metadata timescale (10 seeds). Left: r_3 MC — esn-dual stays 0.68–0.70 below esn-fast at every τ_m , while the homeostat-regulated substrate (redem-reg, no metadata) sits at 8.47. Right: r_3 NMSE — the readout-noise attenuation transfers at every τ_m .

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- [10] Author’s companion Paper B: REDEM: Training-Inference Unified Learning with Meta-Adaptation and Structural Plasticity for Non-Stationary Environments (online learning architecture; target: *Neural Networks*).